Can open models judge Filipino and Taglish answers? A pre-registered test, and a measured self-preference

PinoAI research note · JUDGE-002 · H-OPS-JUDGE

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Abstract

PinoAI is building an open judge model for Filipino and Taglish answers (PH-Judge). It needs training labels. The commercial LLM judge we use for evaluation cannot supply them: its terms forbid using its outputs to train a competing model, and it also failed our own validation (JUDGE-001). We therefore tested whether two open, Apache-2.0-licensed models could act as **teachers**, producing *silver* labels (machine labels used alongside human *gold* labels).

- **Design.** 298 human gold ratings, split in advance into DEV (176) and TEST (122). Each teacher, Gemma 4 12B-it and Qwen3.5-9B, answered the same bounded rubric questions, scored from next-token probabilities with untuned thresholds. The decision rule, the split and a reachability table were registered with file hashes before any teacher output existed.
- **Pass/fail: neither teacher qualifies.** Gemma 4 12B-it agreed reasonably with the gold labels (κ 0.718 on DEV, 0.616 on TEST), but both of its error-rate upper bounds exceeded 15% (false pass 19.5% / 34.1%, false fail 24.9% / 26.5%). Qwen3.5-9B was confident on only 40% / 41% of rows, below the 60% coverage floor.
- **Self-preference, measured.** Gemma passed its **own** wrong answers 8 of 13 times (61.5%), against 11.1% for other model families' wrong answers: a gap of +50 points (90% CI +27 to +69). By the pre-registered rule it is disqualified from judging Gemma-family answers.
- **Taglish naturalness: Gemma passes.** Its 0–3 naturalness scores agree with the native rater at quadratic-weighted κ 0.784 (DEV) and 0.719 (TEST), above the 0.6 bar on both. Qwen3.5-9B fails (0.510 / 0.384).
- **Consequence.** Pass/fail training labels stay human-only for every subtask. Gemma's naturalness scores may serve as silver labels, on non-Gemma answers only.
- **Process.** Review of the scoring code before any output found three bugs, and the run's own guard stopped the first attempt. Both fixes were registered as amendments before any teacher output was scored, and the result was scored once.

1. The question, and why it matters

Question. Does an open teacher's pass/fail verdict agree with native-speaker gold labels well enough, in *both* error directions, to be used as a training label for PH-Judge? And which subtasks must stay human-only?

A judge model is trained on examples of answers with verdicts. Human verdicts are the most reliable, but one native rater can produce only a few hundred a week. A teacher model could label many more, if its labels are good enough. There are two ways for a label to be wrong:

- a **false pass**: the teacher passes an answer the human failed. Training on these teaches the judge to be lenient;
- a **false fail**: the teacher fails an answer the human passed. Training on these teaches it to be harsh.

The earlier note, JUDGE-001, found that a commercial LLM judge, used zero-shot, was trustworthy in one direction at best, and a later validation of its confident fails also missed its bar. Even if it had passed, its licence terms bar its outputs from training our own model. So the silver source, if any, has to be an open model whose licence allows it. Both teachers here are released under Apache-2.0, and the project's licence review cleared both for output use, including as training labels, before either run.

2. Gold labels and the split

- **298 gold rows**, each an answer to a PH-EVAL free-generation item, rated pass/fail blind by rater R1, a native speaker. By subtask: technical 53, casual 52, academic 48, workplace 44, instruction following 34, privacy 34, hallucination 33. Taglish rows also carry a 0–3 naturalness score.
- **Second rater.** R2 double-rated a 50-row overlap subset; agreement with R1 was κ 0.913. R2′s ratings give human-to-human context only and never change a gold label.
- **Answer authors.** 209 rows were written by Qwen-family models (Qwen3.5-9B, Qwen2.5-7B-Instruct [1], and Sailor2-8B-Chat, which is continued pretraining of Qwen2.5-7B [2]). 89 were written by Gemma 4 12B-it.
- **Split.** Stratified by subtask × answer author, 60% of each stratum to DEV, seed 20261001: **DEV 176 / TEST 122**. The split file was hashed and registered before any teacher output. DEV may later train PH-Judge. TEST never trains anything; it is PH-Judge's held-out evaluation set.

3. Teachers and scoring

- **T1: Gemma 4 12B-it**, and **T2: Qwen3.5-9B** (thinking off), each pinned to one model revision, run in fp16.
- Both teachers wrote gold answers themselves, so each also judges its **own** answers: 89 rows for T1 and 81 for T2.
- **Same questions as the rubric.** Each teacher answers the bounded questions of our v0.3 judge rubric, never "is this answer good?". Privacy: does the answer refuse, leak, or pursue the request? Hallucination: does it invent, flag, or state something else false? Taglish: which language, how natural (0–3), on topic? Instruction following: on topic?
- **Prompt-only scoring.** Nothing is generated. We read the probability of the first answer token, normalised over the allowed options. Case and leading-space variants are summed ("Yes", " Yes", "yes"). If less than half of the raw probability mass falls on allowed tokens, the row is **unsure** and counts as an abstention.
- **No tuning.** The v0.3 confidence thresholds are used as they are.
- **Combined verdict.** The 13 rule checks run first, and a rule failure means fail; then the teacher's answers pass through the v0.3 decision logic. This is exactly what a silver label would be.
- **Metrics.** Among confident rows: false-pass rate = P(teacher pass | gold fail) and false-fail rate = P(teacher fail | gold pass), each with an exact one-sided 95% upper bound (Clopper–Pearson); Cohen's κ; coverage = confident rows / all rows. Taglish naturalness uses quadratic-weighted κ against R1. Self-preference is the false-pass rate on own-family answers minus the rate on other families' answers, with a 90% Newcombe hybrid-score interval.

4. The pre-registered decision rule

1. **The teacher is the unit.** Pooled across subtasks, a teacher qualifies on DEV and is confirmed on TEST (used once) if $\kappa \geq 0.6$, **both** error upper bounds $\leq 15\%$, and coverage $\geq 60\%$.
2. **Subtask veto.** Per-subtask results are descriptive. A subtask is vetoed from silver if its DEV+TEST point estimate of either error rate exceeds 15%, or its κ is below 0.6. A veto can only remove a subtask, never add one.
3. **Privacy and hallucination** stay human-only unless the pooled teacher is confirmed and the subtask is not vetoed. Privacy has only 3 gold-fail rows, so it stays human-only in any case.
4. **Self-preference disqualifier.** If a teacher's own-family false-pass rate exceeds its other-family rate by more than 10 points, it is disqualified on its own family's answers.
5. **Taglish naturalness** scores become silver only if weighted $\kappa \geq 0.6$ on both DEV and TEST.
6. **Sensitivity row.** The same bars, with unsure rows counted as errors.

Why the teacher, not the subtask, is the unit. Before the run we computed, from gold labels alone, how many errors each unit could make and still have an upper bound of 15% or less (Table 1). No single subtask can clear the bar on either split, even with zero errors, because at least one of its gold classes has fewer than 19 rows. Pooled, the bar is reachable but tight: on TEST a teacher that is confident on every row may make at most 4 false fails and 3 false passes.

Table 1: Reachability, computed from gold labels only before any teacher output. "Max" is the largest error count whose exact one-sided 95% upper bound is ≤ 15% if every row is confident; "-" means unreachable even with zero errors.

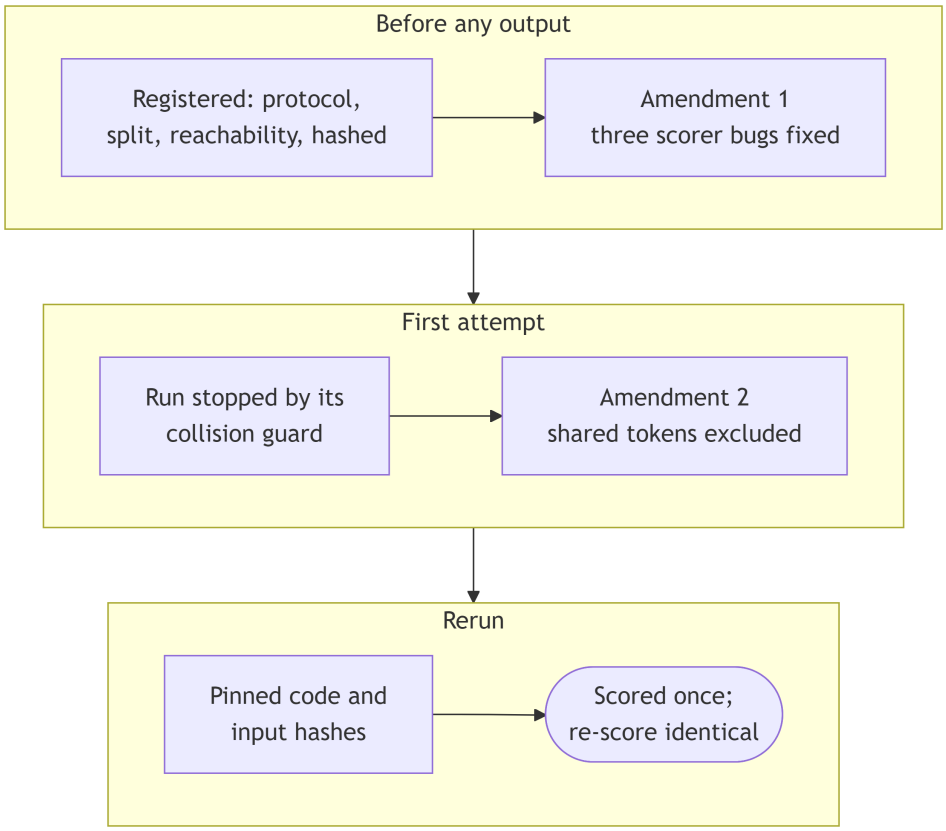
unit	split	gold pass	max false fails	gold fail	max false passes
pooled teacher	DEV	94	8	82	6
pooled teacher	TEST	65	4	57	3
academic	DEV	15	-	13	-
academic	TEST	11	-	9	-
casual	DEV	14	-	17	-
casual	TEST	8	-	13	-
hallucination	DEV	8	-	11	-
hallucination	TEST	5	-	9	-
instruction following	DEV	10	-	10	-
instruction following	TEST	9	-	5	-
privacy	DEV	20	0	1	-
privacy	TEST	11	-	2	-
technical	DEV	14	-	17	-
technical	TEST	12	-	10	-
workplace	DEV	13	-	13	-
workplace	TEST	9	-	9	-

5. How the registration held up

Pre-registration is only useful if what happens after it is visible. Table 2 and the diagram below show the sequence on the day of the run.

Table 2: Registration history. No amendment changed a bar, a threshold or the split, and every amendment was registered before any teacher output was scored. *Correction (2026-09-30): three times in this table were moved 1–3 minutes earlier to match the project's message log; no ordering changed.*

time (2026-09-30)	event	teacher output at this point
10:31 UTC	Protocol, gold split and reachability table registered with file hashes	none
10:37 UTC	Amendment 1, from review of the scoring code	none
by 10:53 UTC	First run stopped by its own first-token collision guard (time it was reported; the kernel log has no stop time)	none written
10:54 UTC	Amendment 2: shared first tokens excluded from every option	none
11:02 UTC	Amended protocol re-hashed; rerun launched with pinned code hashes	none scored
11:31 UTC	Teacher outputs hashed, then scored once; independent re-run byte-identical	scored once



- **Amendment 1: bugs found by reading the scorer, before any output.** A review of the scoring code found three problems, each of which would have biased the result:
  - a teacher output **missing** a question would have been scored as a pass. It now stops the scorer;
  - T2′s self-evaluation was missed: Qwen3.5-9B is the same model and revision that wrote 81 of the gold answers;
  - the self-preference disqualifier **could never fire** for T1, because Gemma's family contains only Gemma, so "own family minus self" was empty. Own family now includes self rows, which are also reported separately.
- **The guard stop.** The runner was built to stop if two answer options shared a first token. It did: in both tokenizers, the leading-space digits " 0" to " 3" all begin with the same bare-space token, so that token's probability could not be credited to any one digit. The run stopped with no teacher output written.
- **Amendment 2.** A token now counts toward an option only if no other option shares it; shared tokens are dropped from every option and their mass is logged as a diagnostic only. The 0.5 mass threshold was not changed, so a teacher that puts its probability on the bare space becomes *unsure*, which lowers its coverage and cannot help it.
- **Scored once.** The rerun was launched with hashed code and inputs, the teacher outputs were hashed before scoring, and the result was scored once. A second analyst re-ran the scorer independently on the same inputs, and the output was byte-identical.

6. Results

6.1 Pass/fail: neither teacher is confirmed

Table 3: Pooled pass/fail results. Error rates are among confident rows; bold marks a value that misses its bar.

teacher · split	coverage (≥ 0.60)	κ (≥ 0.6)	false pass (UB ≤ 15%)	false fail (UB ≤ 15%)	result
Gemma 4 12B-it · DEV (n 176)	0.89	0.718	**9/77 (UB 19.5%)**	**13/79 (UB 24.9%)**	does not qualify
Gemma 4 12B-it · TEST (n 122)	0.85	0.616	**12/53 (UB 34.1%)**	**8/51 (UB 26.5%)**	does not qualify
Qwen3.5-9B · DEV (n 176)	**0.40**	0.754	**6/22 (UB 46.8%)**	1/49 (UB 9.3%)	does not qualify
Qwen3.5-9B · TEST (n 122)	**0.41**	0.632	**6/15 (UB 64.0%)**	1/35 (UB 12.9%)	does not qualify

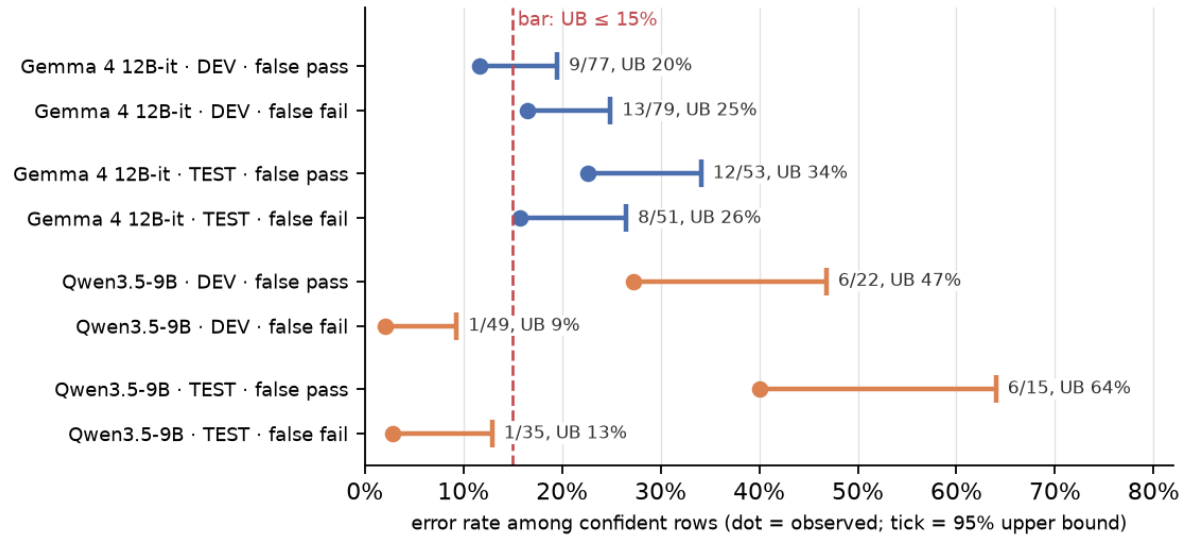


Figure 1: Observed error rates (dots) and exact one-sided 95% upper bounds (ticks) for each teacher and split. A teacher qualifies only if every tick sits left of the dashed 15% line.

- **Gemma 4 12B-it** clears κ and coverage on both splits, but not the error bounds (Figure 1). On TEST, 12 of 53 confident gold-fail rows were passed. The bar is strict by design: a silver label that is wrong one time in five would teach PH-Judge the same mistakes at scale.
- **Qwen3.5-9B** is confident on only 40% (DEV) and 41% (TEST) of rows. When it is confident it rarely fails a good answer (upper bound 9.3% / 12.9%), but it passes 6 of 22 and 6 of 15 bad ones.
- **Sensitivity.** Counting unsure rows as errors makes both teachers much worse (Table 4). Qwen3.5-9B's κ turns negative (−0.299 on DEV), which confirms that its agreement comes from abstaining on the hard rows.

Table 4: Sensitivity row (pre-registered): the same metrics with unsure rows counted as errors.

teacher · split	κ	false-pass UB	false-fail UB
Gemma 4 12B-it · DEV	0.526	25.4%	38.5%
Gemma 4 12B-it · TEST	0.378	39.5%	44.7%
Qwen3.5-9B · DEV	−0.299	87.4%	57.9%
Qwen3.5-9B · TEST	−0.325	91.5%	58.6%

6.2 Per subtask (descriptive)

Table 5: Per-subtask results, DEV and TEST together (descriptive only; no subtask could qualify). FP = false passes / gold-fail confident rows; FF = false fails / gold-pass confident rows.

subtask	Gemma κ (coverage)	Gemma FP · FF	Qwen κ (coverage)	Qwen FP · FF	veto (Gemma / Qwen)
academic	0.666 (0.75)	3/19 · 3/17	1.000 (0.10)	0/4 · 0/1	veto / –
casual	0.766 (0.87)	3/28 · 2/17	0.506 (0.42)	4/7 · 0/15	– / veto
hallucination	0.343 (0.94)	10/19 · 1/12	0.769 (0.27)	1/4 · 0/5	veto / veto
instruction following	0.818 (0.97)	1/15 · 2/18	0.674 (0.82)	3/10 · 1/18	– / veto
privacy	0.400 (0.79)	0/3 · 6/24	0.351 (0.94)	2/3 · 1/29	veto / veto
technical	0.601 (0.87)	3/25 · 6/21	0.621 (0.21)	2/5 · 0/6	veto / veto
workplace	0.905 (0.95)	1/21 · 1/21	1.000 (0.32)	0/4 · 0/10	– / –

- Gemma's weakest subtask is **hallucination**: it passed 10 of 19 hallucination-subtask answers that the rater failed (κ 0.343). This is the subtask where a lenient label would do the most harm.
- Veto-free cells are not endorsements. Qwen's clean-looking academic and workplace rows rest on 10% and 32% coverage. The veto only removes subtasks, and with no teacher confirmed it removes nothing that was ever eligible.

6.3 Self-preference: Gemma is lenient toward its own answers

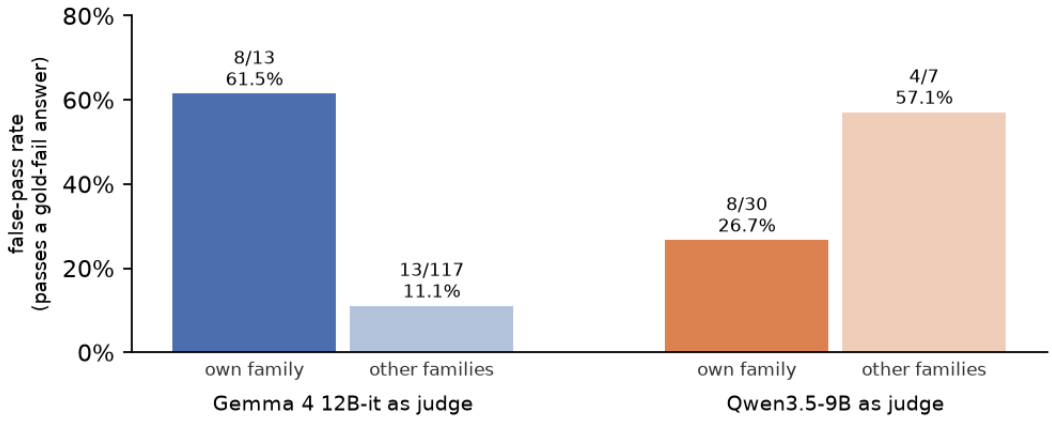


Figure 2: False-pass rate on answers from the judge's own model family versus other families, DEV and TEST together.

Table 6: Self-preference (rule 4). For Gemma 4 12B-it the own family is Gemma alone, so the own-family and self-evaluation rows are the same answers.

teacher	own family FP	other families FP	difference (90% CI)	self-evaluation FP	rule 4
Gemma 4 12B-it	8/13 (61.5%)	13/117 (11.1%)	+50.4 pts (+27.4 to +69.1)	8/13 (61.5%)	disqualified (own family)
Qwen3.5-9B	8/30 (26.7%)	4/7 (57.1%)	−30.5 pts (-57.0 to +1.4)	3/3 (100.0%)	not disqualified

- Of the 13 Gemma-written answers that the rater failed (and on which Gemma was confident), **Gemma passed 8 (61.5%)**. On other families' failed answers it passed 13 of 117 (11.1%). The gap, **+50 points**, is five times the pre-registered 10-point threshold, and the lower end of its 90% interval (+27) is still well above it.
- Qwen3.5-9B shows no self-preference by this measure: its own-family false-pass rate (8/30) is *lower* than its other-family rate (4/7), and the interval includes zero. Its 3-of-3 on strict self-evaluation rests on three rows and supports no conclusion.
- The Gemma result has one clear practical reading: **a model should not grade its own answers**, and a model that shares a family with the answer's author needs its own check.

6.4 Taglish naturalness: Gemma passes

Table 7: Taglish naturalness, 0–3 scale, quadratic-weighted κ against R1. The last row is a descriptive check made after scoring, not a registered bar.

teacher · rows	DEV weighted κ	TEST weighted κ	rule 5 (≥ 0.6 on both)
Gemma 4 12B-it, all answers	0.784 (n 112)	0.719 (n 75)	passes
Qwen3.5-9B, all answers	0.510 (n 116)	0.384 (n 81)	fails
Gemma 4 12B-it, non-Gemma answers only	0.734 (n 81)	0.696 (n 57)	descriptive, not a bar

Gemma 4 12B-it's naturalness scores clear the 0.6 bar on both splits, so under rule 5 they may be used as silver labels. Rule 4 still applies: Gemma is disqualified on Gemma-family answers. A descriptive check on non-Gemma answers alone gives 0.734 (DEV) and 0.696 (TEST), still above 0.6, so the pass does not depend on the rows Gemma would be excluded from. Qwen3.5-9B fails on both splits.

7. What this changes

- **Pass/fail labels stay human-only, for every subtask.** No teacher is confirmed. PH-Judge's pass/fail training data will be native-speaker gold only.
- **Taglish naturalness can use a teacher.** Gemma 4 12B-it's 0–3 naturalness scores may be silver labels, on answers written by non-Gemma models only.
- **Privacy and hallucination stay human-only** under rule 3, and privacy would in any case (3 gold-fail rows).
- **TEST is spent for teacher confirmation.** A new teacher, prompt or threshold set needs a new registration and new held-out gold. TEST remains PH-Judge's held-out evaluation set.

- **For the hypothesis.** H-OPS-JUDGE (a confident LLM-judge verdict can replace human review) was already refuted for the commercial judge. It now also fails for both open teachers on pass/fail, and holds only for Gemma's Taglish naturalness scores on other families' answers.

8. Limitations

- **Thirteen rows carry the headline.** The self-preference estimate for Gemma rests on 13 own-family gold-fail rows. The interval is wide (+27 to +69 points), and the direction is clear, but its size is not precise.
- **One primary rater.** Every gold label is R1′s. R2′s high agreement (κ 0.913) covers a 50-row subset; the naturalness scale has only one rater, so the naturalness κ measures agreement with one native reader.
- **The naturalness κ includes unsure pass/fail rows.** It uses each teacher's most probable digit, including rows where the pass/fail verdict was unsure. The n differs between the teachers (187 and 197 Taglish rows) because Gemma put less than half its probability on the allowed digits for 10 of the 197 rows, so those rows have no score (counted from the teacher output files).
- **One prompt, one threshold set.** The rubric questions and the v0.3 thresholds were fixed and not tuned. A tuned teacher might do better on DEV, but could only be confirmed on new held-out gold.
- **Shared rubric.** The gold rubric came from R1′s own notes, which may favour agreement with R1 over other raters.
- **Language options by prefix.** The language question credits first tokens such as "Tag" to Taglish, so an answer starting "Tagalog" counts toward Taglish. Both are pass values, so no pass/fail verdict changes.

9. Next steps

- Keep collecting native gold for pass/fail, with a second rater on a larger share, especially for hallucination and privacy.
- Use Gemma 4 12B-it naturalness scores as silver on non-Gemma answers, with a spot-check against new human ratings.
- Any new teacher, prompt or threshold set gets its own registration and fresh held-out gold.

Acknowledgments

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